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# A Landsat time series approach to characterize bark beetle and defoliator impacts on tree mortality and surface fuels in conifer forests

Garrett W. Meigs <sup>a,\*</sup>, Robert E. Kennedy <sup>a</sup>, Warren B. Cohen <sup>b</sup>

<sup>a</sup> Department of Forest Ecosystems and Society, Oregon State University, 321 Richardson Hall, Corvallis, OR, 97331, USA

<sup>b</sup> Pacific Northwest Research Station, USDA Forest Service, 3200 SW Jefferson Way, Corvallis, OR, 97331, USA

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### ABSTRACT

Insects are important forest disturbance agents, and mapping their effects on tree mortality and surface fuels represents a critical research challenge. Although various remote sensing approaches have been developed to monitor insect impacts, most studies have focused on single insect agents or single locations and have not related observed changes to ground-based measurements. This study presents a remote sensing framework to (1) characterize spectral trajectories associated with insect activity of varying duration and severity and (2) relate those trajectories to ground-based measurements of tree mortality and surface fuels in the Cascade Range, Oregon, USA. We leverage a Landsat time series change detection algorithm (LandTrendr), annual forest health aerial detection surveys (ADS), and field measurements to investigate two study landscapes.

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